

GRADE 6 | DAY 5

MONDAY, AUGUST 31, 2026

START TIME: 9:00 AM

TODAY'S GOAL

Begin Week 2 by using precise vocabulary, showing reasoning, and moving forward where Friday showed readiness.

Lesson order is intentionally different from Friday.

Friday was a partial instructional day because of a doctor appointment. Unattempted Friday sections are not missing work.

Time	Lesson
9:00-9:45	Science - Transparent, translucent, opaque + scattering repair
9:50-10:25	French - Etre and avoir + negation review
10:30-11:30	Mathematics - Ratios: comparing quantities
11:35-12:15	History/Geography - Athens and Sparta: two poleis
12:15-1:00	Lunch
1:00-1:50	Literature - Fiction: inference, character, theme
1:55-2:35	Writing - Sentence boundaries + evidence-based explanation
2:40-3:20	Computer Science / AI - Confidence, correctness, generalization
3:25-3:55	Art / Design - Friday portfolio carryover
3:55-4:05	Closeout - Started -> Finished -> Checked -> Filed

9:00

WEEK 1 FEEDBACK BEFORE MONDAY BEGINS

Read this once. Then start Science.

WHAT CHANGED FRIDAY

Mathematics: decimal division was strong across guided practice, independent work, applications, and the exit ticket. Monday moves into ratios, with a short decimal-division warmup.

Literature: topic and summary precision improved substantially on the bridge article. Monday switches to fiction so you practice inference, character, and theme.

Writing: the bridge paragraph used accurate facts and real reasoning. The next priority is sentence boundaries and punctuation so strong ideas are easier to read.

Science: the light-path idea is strong, but transparent / translucent / opaque and scattering are still mixed up. Monday starts by repairing those exact terms.

History: the geography -> polis cause/effect idea is strong enough to move into Athens and Sparta.

French: regular -ER verbs and ne...pas are improving. Monday adds two high-frequency irregular verbs: *être* and *avoir*.

Computer Science / AI: you understand that machine learning learns from examples. Monday sharpens what model confidence means - and what it does NOT mean.

QUESTION-QUALITY RULE

Every question today names exactly what it is asking about.

No vague prompts such as 'Who wrote the second condition?' If a question refers to a rule, passage, model, or sentence, the exact item is named in the question.

9:00-9:45

SCIENCE 6 - LIGHT THROUGH MATERIALS

Repair classification using transmission + image clarity.

WORDS YOU NEED TODAY

transmit - let light pass through

scatter - redirect light in many directions

transparent - lets most visible light through with little scattering, so a clear image can form

translucent - lets some light through but scatters it, so the image is blurry or unclear

opaque - does not let visible light pass through

evidence - an observation or measurement used to support an explanation

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus the three material categories using TWO tests: (1) does visible light get through, and (2) does a clear image form? Start with his Friday evidence: clear plastic allowed a clear view; wax paper allowed light through but the image was not clear; cardboard blocked the view. Explain that 'light gets through' alone does not make something transparent. Model three examples before practice. Ask self-contained questions such as: 'Wax paper lets light through but blurs the object. Which category fits, and what observation supports it?' Do not ask vague referential questions.

Teaching: use two questions, not one

1. Does visible light pass through the material?
2. If it passes through, does a clear image form?

WORKED EXAMPLE 1

Clear window glass: light passes through and objects stay clear -> transparent.

WORKED EXAMPLE 2

Wax paper: light passes through, but details look blurry because light is scattered -> translucent.

WORKED EXAMPLE 3

Cardboard: visible light does not pass through -> opaque.

Guided practice

1. Frosted bathroom glass lets light through, but you cannot see a sharp image through it. Is it transparent, translucent, or opaque? Explain using both transmission and image clarity.

2. A black sheet of poster board blocks a flashlight beam. Which category fits? What observation is your evidence?

3. A clear plastic ruler lets you read printed words sharply through it. Which category fits, and why?

9:00-9:45

SCIENCE 6 - INDEPENDENT + EXIT

Use observations, not memorized labels.

1. A lampshade glows when the lamp is on, but you cannot see the bulb clearly through the shade. Classify the lampshade material and explain.

2. Explain the difference between 'transmits light' and 'forms a clear image.'

3. Why can a translucent material make a room brighter even when you cannot see a clear object through it?

4. A material reflects some light and transmits some light. Does that fact alone tell you whether it is transparent or translucent? Explain what additional observation you need.

EXIT TICKET - NO HELP

E1. Wax paper lets visible light through but makes printed letters look blurry. Name the material class and explain why.

E2. In one sentence, explain what scattering does to the direction of light.

9:50-10:25

FRENCH 6 - ETRE AND AVOIR

New irregular verbs + negation review.

WORDS YOU NEED TODAY

etre - to be

avoir - to have

irregular verb - a verb that does not follow the normal regular -ER pattern

conjugate - change a verb form to match the subject

negation - a form such as ne...pas that makes a sentence negative

elision - dropping the vowel in ne before a vowel sound: n'ai, n'est

LISTEN FIRST - VOICE TUTOR PROMPT

Teach continuing French, not beginner French. Review one ne...pas sentence, then introduce the present tense of etre and avoir. Speak the French forms naturally and have Atticus repeat them. Explain n'ai and n'est as elision before vowel sounds. Use short exchanges: 'Tu as un livre?' 'Oui, j'ai un livre.' 'Tu es fatigued?' 'Non, je ne suis pas fatigued.' Correct only high-value errors. Questions must name the exact verb or sentence being changed.

Forms to know

ETRE: je suis, tu es, il/elle est, nous sommes, vous etes, ils/elles sont

AVOIR: j'ai, tu as, il/elle a, nous avons, vous avez, ils/elles ont

Worked examples

EXAMPLE 1

Je suis pret. -> Je ne suis pas pret.

EXAMPLE 2

J'ai un livre. -> Je n'ai pas de livre.

EXAMPLE 3

Elle est fatiguée. -> Elle n'est pas fatiguée.

Practice

1. Complete with *être*: Nous _____ prêts.

2. Complete with *avoir*: Ils _____ deux livres.

3. Make this sentence negative: J'ai un crayon.

4. Make this sentence negative: Elle est contente.

5. Answer in a complete French sentence: Est-ce que tu as un livre aujourd'hui?

6. Answer in a complete French sentence: Est-ce que tu es fatigué ce matin?

ORAL CHECK

Say four short sentences using at least two forms of *être* and two forms of *avoir*. Include one negative sentence.

10:30-11:30

MATHEMATICS 6 - RATIOS

New Grade 6 unit after strong decimal-division evidence.

WORDS YOU NEED TODAY

ratio - a comparison of two quantities by division

equivalent ratios - ratios that describe the same comparison

part-to-part ratio - a comparison of one part of a group with another part

part-to-whole ratio - a comparison of one part with the entire group

scale factor - the number used to multiply or divide both quantities in a ratio

ratio table - a table showing pairs of equivalent ratio values

LISTEN FIRST - VOICE TUTOR PROMPT

Begin with a 5-minute decimal-division warmup because Friday showed strong performance, then introduce ratios as a new Grade 6 unit. Make clear that a ratio compares two quantities in a stated order. Model three forms: 3 to 5, 3:5, and $\frac{3}{5}$. Explicitly distinguish part-to-part from part-to-whole. Then model equivalent ratios by multiplying BOTH quantities by the same scale factor. Ask clear questions that name both quantities; never use an unexplained 'first' or 'second' condition.

Warmup - keep decimal division alive

1. $6.3 \div 0.9 =$

2. $4.8 \div 0.6 =$

Teaching

A ratio tells how much of one quantity there is compared with another quantity. ORDER MATTERS.

WORKED EXAMPLE 1

A box has 3 blue markers and 5 black markers. Blue to black = 3:5.

WORKED EXAMPLE 2

The same box has 8 markers total. Blue to total = 3:8. This is part-to-whole, not 3:5.

WORKED EXAMPLE 3

If 2 cups of concentrate mix with 5 cups of water, then 4 cups of concentrate mix with 10 cups of water. 2:5 and 4:10 are equivalent because both quantities were multiplied by 2.

Guided practice

1. A tray has 4 red tiles and 7 white tiles. Write the ratio of red tiles to white tiles in three forms.

2. The same tray has 11 tiles total. Write the ratio of red tiles to all tiles.

3. Complete the equivalent ratio: $3:8 = 6:\underline{\hspace{1cm}}$. State the scale factor.

4. Complete the ratio table for 2 notebooks per 5 pencils: (2,5), (4,__), (6,__), (__,20).

10:30-11:30

MATHEMATICS 6 - INDEPENDENT + EXIT

State the quantities in the order asked.

1. A model kit contains 6 metal pieces and 9 plastic pieces. Write the ratio of metal pieces to plastic pieces in simplest form.

2. The same kit has 15 pieces total. Write the ratio of plastic pieces to all pieces in simplest form.

3. A paint mixture uses 3 parts blue for every 4 parts white. If 12 parts white are used, how many parts blue are needed? Show an equivalent-ratio method.

4. A snack mix uses 5 cups of cereal for every 2 cups of nuts. Complete: 10 cups cereal -> ____ cups nuts; 15 cups cereal -> ____ cups nuts.

5. Are 4:6 and 10:15 equivalent ratios? Show how you know.

EXIT TICKET - NO HELP

E1. A shelf has 8 fiction books and 12 nonfiction books. What is the ratio of fiction to nonfiction in simplest form?

E2. Write one ratio that is equivalent to 5:7 and explain the scale factor.

E3. Explain why the ratios 2:3 and 4:5 are NOT equivalent.

11:35-12:15

HISTORY & GEOGRAPHY - ATHENS AND SPARTA

Two Greek poleis with different priorities.

WORDS YOU NEED TODAY

Athens - a Greek polis known for trade, public debate, arts, and an early form of democracy for male citizens

Sparta - a Greek polis known for military organization and strict social discipline

citizen - a recognized member of a political community with certain rights and duties

democracy - government in which citizens participate in political decisions

oligarchy - government controlled by a small group

compare - identify similarities

contrast - identify differences

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus that Athens and Sparta were both Greek poleis but developed different political and social systems. Give geographic context first. Make clear that ancient Athenian democracy excluded women, enslaved people, and many residents, so it was not modern universal democracy. Ask questions with explicit subjects: 'How did Athens differ from Sparta in government?' rather than 'How were they different?' without a referent.

Reading - Two Poleis, Different Priorities

Athens and Sparta were two influential Greek poleis. Both spoke Greek, worshiped many of the same gods, and participated in shared festivals, but they organized society differently.

Athens developed near the sea and became active in trade. Public speaking, law, drama, philosophy, and civic debate became important parts of Athenian life. Adult male citizens could participate directly in parts of government. This system is often called an early democracy, although most people living in Athens did not have full political rights.

Sparta developed a society centered more heavily on military readiness and discipline. A small group of Spartan citizens trained for warfare, while other groups performed much of the farming and labor. Sparta had kings, councils, and assemblies, but political power was concentrated in fewer hands than in

Athens.

The contrast shows why the Greek world was not one uniform society. Poleis shared cultural traditions while making different choices about government, education, military life, and public participation.

1. Name TWO cultural features Athens and Sparta shared according to the reading.

2. How did government participation in Athens differ from political power in Sparta?

3. Why is it inaccurate to say that ancient Athenian democracy gave equal political rights to everyone living in Athens?

4. What larger idea about the Greek world is shown by the differences between Athens and Sparta?

EXIT TICKET

In 3-4 sentences, compare Athens and Sparta. Include one similarity, one difference, and one sentence explaining how the comparison supports the idea that Greek poleis were culturally connected but politically independent.

1:00-1:50

LITERATURE - FICTION CLOSE READING

Inference, character motivation, and theme.

WORDS YOU NEED TODAY

character motivation - the reason a character chooses to act

inference - a conclusion based on text evidence plus reasoning

conflict - a struggle or problem that drives a story

theme - a complete message about life, people, or choices developed by a story

evidence - a specific detail from the text that supports an interpretation

objective summary - a brief account of the main events without opinion

LISTEN FIRST - VOICE TUTOR PROMPT

Monday switches from informational text to fiction. Teach Atticus how to infer motivation and theme from actions and consequences. Do not give the theme before he reads. Model one unrelated example: a character hides a broken tool, then later admits it and helps repair the damage; evidence can support a theme about responsibility and trust. After Atticus reads, ask exactly which action or line supports his inference. Keep theme universal, not 'the character learned...'.

Original story - The Spare Key

By the time Malik reached the makerspace, the rain had already turned the windows gray. His team had one hour left to test the small weather station they had built for the school courtyard.

The station was supposed to record temperature and rainfall. On Friday, Malik had volunteered to carry the sensor case home so he could tighten a loose bracket. He had also promised to bring it back first thing Monday.

When he opened his backpack, the case was there. The tiny brass key that opened it was not.

Malik checked every pocket twice. Across the room, Priya was connecting wires while Jonah loaded the data program. 'We should start the sensor test,' Priya said. Malik felt his face grow warm.

He could tell them the lock had jammed. That would buy time. Maybe the key was in his room, under a notebook or beside the charger. If he stayed quiet, no one would know he had misplaced it.

Then Jonah held up the test checklist. 'Step one says inspect the sensor before power-on. We can't do that with the case locked.'

Malik looked at the case and then at the clock. 'I lost the key,' he said. 'I think it's at home. I should've checked before I left.'

No one cheered. Priya closed her eyes for a second. Then she opened the supply cabinet. 'There might be a spare.'

They searched through labeled drawers until Jonah found a small envelope taped behind the cabinet door: WEATHER STATION - SPARE KEY. The test began twenty minutes late.

During the first trial, the rainfall sensor gave impossible numbers. Because the case was open, Malik noticed that one wire had loosened during the trip home. He reconnected it, and the second trial worked.

At cleanup, Malik wrote a new line on the checklist: RETURN KEY TO LAB BEFORE TAKING CASE HOME. Priya added another: CHECK SPARE BEFORE TEST DAY.

'So we're making a rule because you lost a key?' Jonah asked.

Malik shook his head. 'We're making a better system because one person shouldn't be able to stop the whole test.'

Priya nodded. 'That I can use.'

1:00-1:50

LITERATURE - CLOSE READING QUESTIONS

Use evidence when the question asks for it.

1. What is Malik's main problem at the beginning of the story?

2. What does Malik consider doing instead of admitting he lost the key? What does that reveal about his immediate motivation?

3. Why is Malik's decision to tell the truth important to the plot? Use one specific event that happens afterward.

4. What can you infer about Priya from the way she responds after Malik admits the problem? Use one detail.

5. At the end, Malik says, 'one person shouldn't be able to stop the whole test.' What larger lesson has he drawn from the mistake?

6. Write a universal theme statement for the story. Do not use a character's name in the theme.

7. Give TWO pieces of evidence that support your theme.

8. Write a 3-sentence objective summary of the story.

EXIT TICKET

What is the difference between a topic such as 'mistakes' and a theme about mistakes? Answer in 1-2 sentences.

1:55-2:35

WRITING - SENTENCE BOUNDARIES

Keep strong ideas; make each sentence easier to follow.

WORDS YOU NEED TODAY

complete sentence - a group of words with a complete thought

run-on sentence - two or more complete thoughts joined without correct punctuation or a connecting word

comma splice - two complete sentences joined only with a comma

conjunction - a connecting word such as and, but, so, or because

revise - improve ideas or sentence clarity

edit - correct punctuation, capitalization, spelling, or grammar

LISTEN FIRST - VOICE TUTOR PROMPT

Use Friday's writing strength as the starting point: his content and evidence were good. Teach sentence boundaries without rewriting his whole paragraph. Explain three repair choices: period, semicolon when appropriate, or comma + conjunction. Model three examples. Then have Atticus repair sentences independently and write a short explanation. Questions must identify the exact sentence being revised.

Worked examples

EXAMPLE 1 - RUN-ON

Too long: The metal warms it expands the joint gives it room to move.

Repair: The metal warms and expands. The joint gives it room to move.

EXAMPLE 2 - COMMA SPLICE

Incorrect: The model was confident, it was still wrong.

Repair: The model was confident, but it was still wrong.

EXAMPLE 3 - BECAUSE

Clear: Engineers add joints because blocked expansion can create damaging stress.

Independent sentence repair

1. Repair this run-on: 'The bridge gets warmer the metal expands the sections need room to move.'

2. Repair this comma splice: 'The classifier showed 99% confidence, the prediction was incorrect.'

3. Combine these ideas with because: 'Wax paper is translucent. It transmits light but scatters it.'

4. Write one sentence explaining why punctuation can change how easily a reader understands an idea.

Short writing

Write 6-8 sentences explaining: Why is a good system designed so that one person's mistake does not automatically cause the whole process to fail? Use the story 'The Spare Key' as one example and include at least one clear cause/effect connection.

WORDS YOU NEED TODAY

prediction - the class or output a model chooses for an input

confidence - a number showing how strongly the model matches an input to one of its learned classes

correctness - whether the prediction is actually right

generalization - working well on new examples not seen exactly during training

training data - examples used to teach a model

bias - a systematic tendency caused by data, design, or assumptions that can make performance uneven

LISTEN FIRST - VOICE TUTOR PROMPT

Build directly from Friday's Teachable Machine work. Atticus already knows that a model can learn from images and can generalize to new examples. Now separate confidence from correctness. Use this exact example: a classifier labels a book as 'cup' with 99% confidence. Explain that 99% describes the model's internal match among the classes it knows; it does NOT prove the object is a cup or guarantee a 99% chance of real-world correctness. Then use a short Teachable Machine test if time allows. Ask concrete questions naming the prediction, confidence value, and actual object.

Think before testing

1. A classifier sees a BOOK but predicts CUP with 99% confidence. Is the prediction correct? What does the 99% tell you?

2. Why might a classifier trained on cups photographed only on a white table perform worse when the cup is photographed outdoors?

3. Which training set is more likely to generalize: 30 nearly identical cup photos, or 30 cup photos with varied angles, distances, and backgrounds? Explain.

Optional online test - Teachable Machine

Open: <https://teachablemachine.withgoogle.com/>

Use household objects only. No faces or personal information.

Train two simple classes with at least 15 examples each. Change ONE condition - angle, distance, background, or lighting - and record what happens.

4. Name the one condition you changed.

5. Before the change, what did the model predict and what confidence did it show?

6. After the change, what did the model predict and what confidence did it show?

7. Did the result show strong or weak generalization? Explain using the before/after evidence.

EXIT TICKET

E1. Explain in one sentence why high model confidence does not guarantee a correct prediction.

E2. Name one change to training data that could improve generalization.

3:25-3:55

ART / DESIGN - PORTFOLIO CARRYOVER

Friday's missed block: complete it today with no penalty.

WORDS YOU NEED TODAY

visual hierarchy - the order in which a viewer notices information

annotation - a short label or note explaining part of a visual

contrast - a difference in size, weight, shape, or tone that helps an important element stand out

composition - how visual elements are arranged on the page

caption - brief text that explains the visual

LISTEN FIRST - VOICE TUTOR PROMPT

This is a carryover from Friday's shortened day, not a missing-work penalty. Help Atticus make a one-page visual explainer of ONE Week 1 concept. Ask which concept he wants to teach, who the imagined viewer is, and what the viewer should notice first. Require a clear title, one main diagram, at least three useful annotations, deliberate hierarchy, and a 2-3 sentence caption. Do not design it for him.

Choose ONE concept

- Ratios: part-to-part vs part-to-whole
- Transparent vs translucent vs opaque
- Athens vs Sparta
- Etre and avoir
- Confidence vs correctness in machine learning
- Why bridge expansion joints matter

What should the viewer notice FIRST?

What is the main diagram or visual?

List at least THREE labels or annotations you will include.

Draft your 2-3 sentence caption.

FINAL REQUIREMENTS

Clear title

One central visual

At least 3 useful labels

One deliberate use of size/contrast for hierarchy

2-3 sentence accurate caption

Name + date

3:55-4:05

MONDAY CLOSEOUT

Started -> Finished -> Checked -> Filed

NOTHING MISSING CHECK

- Science independent + exit ticket
- French practice + oral check
- Math independent + exit ticket
- History questions + exit
- Literature close reading + exit
- Writing sentence repair + short explanation
- CS/AI reasoning + exit ticket
- Art/Design visual explainer

One idea I can explain better now:

One word from today's glossary I can define without looking:

One question I still have:

Monday glossary - all subjects

Subject	Term	Plain-language meaning
Science	transparent	lets most visible light through with little scattering, allowing a clear image
Science	translucent	lets some visible light through but scatters it, so images are blurry
Science	opaque	does not transmit visible light

Science	scatter	redirect light in many directions
French	etre	to be
French	avoir	to have
French	irregular verb	a verb that does not follow the usual regular pattern
Math	ratio	a comparison of two quantities by division
Math	equivalent ratios	ratios that describe the same comparison
Math	scale factor	the number used to multiply or divide both quantities
History	citizen	a recognized member of a political community with rights and duties
History	democracy	government in which citizens participate in political decisions
History	oligarchy	government controlled by a small group
Literature	motivation	the reason a character chooses to act
Literature	conflict	a struggle or problem that drives a story
Writing	run-on sentence	two or more complete thoughts joined without correct punctuation
Writing	comma splice	two complete sentences joined only with a comma
CS/AI	confidence	how strongly a model matches an input to one of its classes; not a guarantee of correctness
CS/AI	generalization	working well on new examples not seen exactly during training
Art	visual hierarchy	the order in which a viewer notices information

Friday evidence used for Monday

Friday was treated as a partial instructional day because of a doctor appointment. Unattempted Art/Design and Week 1 closeout work were not scored as missing. Decimal division evidence was strong enough to begin ratios. Reading precision improved. Writing content/evidence improved markedly; mechanics remain the focus. Science material-classification vocabulary/scattering remains the clearest reteach need. History and French are ready to progress. CS/AI needs precision on confidence versus correctness.

Answer key

Science guided: 1 translucent; light passes but image not sharp. 2 opaque; no light passes. 3 transparent; light passes and words remain clear.

Science independent: 1 translucent. 2 Transmission asks whether light passes; image clarity asks whether scattering preserves a sharp image. 3 It transmits scattered light. 4 No; observe whether a clear image forms / degree of scattering. Exit: wax paper = translucent; scattering redirects light in many directions.

French: 1 sommes. 2 ont. 3 Je n'ai pas de crayon. 4 Elle n'est pas contente. Open responses vary.

Math warmup: $6.3/0.9=7$; $4.8/0.6=8$.

Math guided: 1 4:7, 4 to 7, 4/7. 2 4:11. 3 16, scale factor 2. 4 (4,10), (6,15), (8,20).

Math independent: 1 2:3. 2 3:5. 3 9 blue parts. 4 4 cups nuts; 6 cups nuts. 5 yes, both simplify to 2:3.

Math exit: E1 2:3. E2 e.g. 10:14, scale factor 2. E3 not equivalent because $2/3 \neq 4/5$.

History: 1 Greek language/religion/festivals etc. 2 Athens allowed direct participation for adult male citizens; Sparta concentrated power more narrowly. 3 Many residents had no political rights. 4 Poleis could share culture while differing politically/socially.

Literature: Key ideas: Malik loses key; considers hiding mistake; admits it; team finds spare; open case reveals loose wire; team improves checklist/system. Theme examples: admitting mistakes and improving systems can rebuild trust; responsible people learn from errors and make future failure less likely.

Writing: Accept any correct sentence-boundary repairs. Require 6-8 complete sentences with a clear explanation, one story example, and cause/effect.

CS/AI: 1 Prediction is wrong; 99% means the model strongly preferred CUP among its known classes, not that reality is 99% certain. 2 narrow background-specific training can overfit. 3 varied set. Exit: confidence $\neq$ correctness; improve variety/coverage/representativeness.

Mastery decision rule after Monday

Math: if ratio exit is 3/3 with explanation quality, continue to ratio tables/unit rates. Science: if both science exit items are correct and explanations use transmission + clarity/scattering accurately, classification can move from developing toward proficient. Writing: grade content/reasoning separately from mechanics. Do not reduce reading difficulty because punctuation needs work.